

Failure Cases

Where our system can break or perform poorly, why, and what we do about it. Figures marked "sim" come from our PyBullet runs (default seed).

1. SLAM heading drift in long, featureless corridors

- **Limitation:** stock BreezySLAM (RMHC scan matching) drifts in heading along long straight corridors, about $-0.2^\circ/\text{m}$ measured. Turning in place makes it worse: -36° over 18 s of turning.
- **Why:** a perfectly featureless straight tunnel gives scan matching nothing to hold on to along the corridor.
- **How we handle it:** damped fusion of scan matching with odometry and gyro, no scan-match correction while rotating fast, and a wall-direction heading reference. We also added wall recesses to the simulated tunnels. Result (sim): mean position error 0.14 m, max 0.28 m, while odometry alone ends 4-8 m off (Figure 1).
- **What remains:** the wall-direction reference assumes an orthogonal, room-and-pillar mine layout. For irregular tunnels we would turn it off and expect gyro-bias heading drift. The map is also centred on the start pose, so the mine must lie within ± 40 m of it.

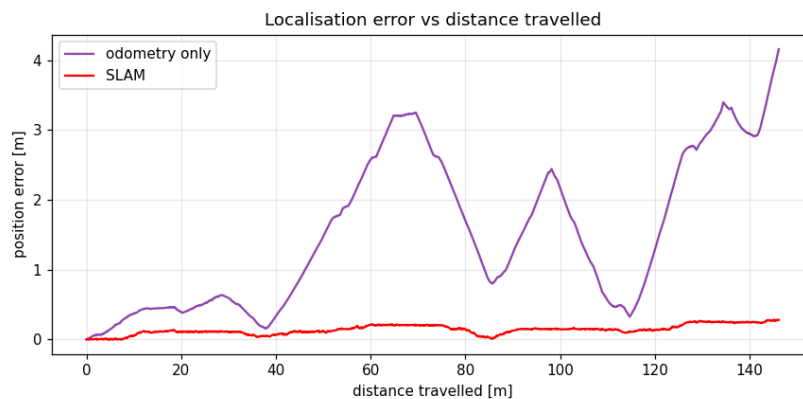


Figure 1. Position error vs distance travelled: SLAM (red) against odometry only (purple).

2. Sensor false positives and false negatives

- **MQ-2 gas:** it is not selective for toxic gases and is affected by temperature and humidity. Dust, exhaust or other airborne chemicals can raise the reading and cause a false alarm. We use it as a threshold-above-baseline gas/smoke anomaly trigger, not a calibrated measurement. The long preheat and baseline drift also matter. Mitigation: trigger only on a sustained rise above the warmed-up baseline, and give gas MEDIUM priority.
- **PIR motion (trapped worker):** it can false-trigger on warm equipment or moving air. It can also miss a worker who lies perfectly still, because it detects warm bodies that *move*. The robot's own movement changes what the PIR sees. Mitigation: ignore triggers during the 30-60 s warm-up, read only while the robot is stopped, and require repeat triggers. A missed still worker is a real risk we cannot remove with this sensor.
- **Tilt/vibration switch (collapse):** the ball bounces and also trips on the robot's own bumps. It cannot tell how big an event was. Mitigation: debounce in firmware, require a few triggers inside a short window, and tune the sensitivity potentiometer on the real chassis.
- **What our simulation does not test:** simulated events fire when the robot's true position enters a predefined zone, so false positives and negatives are never exercised. Bench tests in Phase 2 are where these get measured.

3. RF range limits

- **Limitation:** nRF24L01 radios have limited range, and rock and metal structures underground cut it further. If the Writer gets too far from the outside network node, beacons can be lost.
- **Why it matters:** this limits how large a tunnel network we can cover with a single node.
- **How we handle it:** beacons carry a priority-based TTL and rebroadcast interval (high priority: TTL 20, every 2 s), so urgent findings get more chances to get through. That does not help once the Writer is fully out of range.
- **Known direction (not built):** chain several relay nodes as a mesh, and use PA+LNA radio modules if range tests demand it.

- **Clock assumption:** beacon aging assumes every node shares one mission clock started at deployment. If the clocks drift apart, beacons expire too early or too late. Phase 2 needs a simple clock sync at the start of the mission.
- **What our simulation does not test:** radio delivery is a direct function call in simulation, so packet loss is not modelled. The Phase 2 radio test measures packet loss against distance.

4. Frame translation assumptions

- **Limitation:** the GPS conversion uses a flat-earth approximation and a one-time manual calibration at the tunnel entrance (GPS position and compass bearing of the robot's initial heading).
- **Why:** the approximation is fine at the scale of one tunnel network, but accuracy would degrade over much larger distances. A wrong calibration reading shifts every beacon consistently. A heading error of 2° moves a beacon 100 m from the entrance by about 3.5 m, and 5° by about 8.7 m. A phone GPS fix error moves every beacon by the same offset.
- **How we handle it:** we chose manual calibration over an onboard magnetometer, since compasses are unreliable near metal and mineral content. The Executor uses beacon coordinates in the shared entrance frame, so GPS calibration errors affect the command-post map but not the Executor's driving. Placing both robots at the same marked entrance pose matters, and the Phase 2 plan measures it.

5. Executor navigation: tested only in simulation, with known limits

- **History (what we fixed):** the first Executor controller steered straight at its target and rotated in place when blocked. In the integration demo it reached only 1 of 3 beacons before the 900 s time limit, and its position error grew to 1.88 m. It now plans a path over its own SLAM map with A* (walls inflated by 0.35 m, unknown space allowed but costlier), follows it with pure pursuit, and re-plans every 1.5 s or when blocked.
- **Current result (sim):** with the new controller the Executor visited 3 of 3 beacons on four seeds (default seed 7, plus seeds 1-3), in about 136 s and 63 m, with no wall contacts. (An earlier version brushed the walls 55 times in the default run; keeping 0.45 m clear of walls and a shorter look-ahead fixed it.)
- **Remaining limits:** the planner treats unknown space as passable, so in an unexplored area it may head into a dead end and only re-plan once the LiDAR reveals the wall. Its steering gains are reused from the Writer's wall-following controller and were not separately tuned. It assumes a static tunnel (no moving obstacles or new rock falls), and the same SLAM drift risks from case 1 apply. All results are simulated on one mine layout and have not been tested on hardware; real motor, wheel-slip and LiDAR noise will be worse. Phase 2 repeats the test in a physical mock tunnel.

6. Firmware and hardware are verified in simulation only

- **Limitation:** the ESP32 firmware has not been built with the real toolchain or run on a robot. In simulation, the real firmware code (compiled for a PC) drives a simulated robot through the mine with the real SLAM, controllers, radio medium and command post. The Writer finishes with a mean position error of 0.25 m (maximum 0.45 m); the Executor visits 3 of 3 beacons. The same result holds when 50 % of radio sends fail, because beacons are re-sent until acknowledged.
- **Why it can still fail on hardware:** pin assignments, motor and encoder directions, the tilt-switch polarity, the LD06 CRC and angle direction, wheel slip and radio range underground are all assumptions until bring-up. The simulated sensors have no dust, no radio interference and no MQ-2 drift.
- **How we handle it:** a written bring-up checklist (firmware/README.md), parameters for every value that may need adjusting, a firmware safety layer that stops the robot on command timeout, LiDAR loss or an obstacle, and a Phase 2 plan that tests each part alone before the whole.